Patapanchala Chandra Sekhar

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.15
Sri Chaitanya College(Srichakra Bhavan), Vijayawada	Class XII (Board of Intermediate Education Andhra Pradesh)	2023	95.60%
Dr. KKR's Gowtham Concept School, Gudivada	Class X (Board of Secondary Education Andhra Pradesh)	2021	100.00%

POSITIONS OF RESPONSIBILITY

- **Hospitality and Infra Coordinator** at **MILAN**, The General Championship of IIT Hyderabad
- **Transport Coordinator**, The Student Body of IIT Hyderabad

SKILLS

- **Programming Languages**: C, C++.
- **Technical Skills**: Solid Edge, Solid Works, LaTex, AutoCAD, ABAQUS, Surveying, Soil Testing Methods.
- **Problem Solving :** Data Structures and Algorithms, Object Oriented Programming.
- **Misc :** Git, GitHub.

RELEVANT COURSES

- **Computer Science:** Introduction to Programming
- **Mathematics**: Probability, Introduction to Statistics, Linear Algebra, Calculus, Differential Equations, Transform Techniques, Complex Variables
- **Civil courses:** Construction Materials, Engineering Drawing, Engineering Mechanics, Construction Management, Surveying, Environmental Systems Engineering I & II, Fluid Mechanics, Mechanics of Solids and Structures, Hydraulic Engineering, Geotechnical Engineering, Structural Analysis.
- **Miscellaneous :** Digital Fabrication, Communication Skills, Innovation Management.

PROJECTS

Topology Optimization of Beams

- Performed **Topology Optimization of beams** to achieve lightweight and high-stiffness structural designs under multiple loading and boundary conditions.
- Applied **finite element analysis (FEA)** with compliance and buckling load factor (BLF) constraints to ensure both strength and stability of optimized beam structures.
- Implemented **density and sensitivity filtering techniques** to eliminate numerical instabilities and obtain **mesh-independent solutions**.
- Demonstrated **efficient material distribution** and significant weight reduction in optimized beam designs,

highlighting practical applications in civil, aerospace, and automotive engineering.

EXTRACURRICULAR

- Participated in **TATA Crucible Campus Quiz 2024**, a premier national-level business and knowledge quiz organized by the Tata Group.
- Volunteered in **organizing** and **coordinating** welcome activities for the 2024 batch.
- Active **NSS Volunteer**.
- Achieved **First Place (Gold Medal)** in Departmental Cricket Championship, Highlighting strong team coordination and competitive spirit.